

Thus Mandelbrot's fractals allow us to account for a few Black Swans, but not all. I said earlier that some Black Swans arise because we ignore sources of randomness. Others arise when we overestimate the fractal exponent. A gray swan concerns modelable extreme events, a black swan is about unknown unknowns.

I sat down and discussed this with the great man, and it became, as usual, a linguistic game. In [Chapter 9](#) I presented the distinction economists make between Knightian uncertainty (incomputable) and Knightian risk (computable); this distinction cannot be so original an idea to be absent in our vocabulary, and so we looked for it in French. Mandelbrot mentioned one of his friends and prototypical heroes, the aristocratic mathematician Marcel-Paul Schützenberger, a fine erudite who (like this author) was easily bored and could not work on problems beyond their point of diminishing returns. Schützenberger insisted on the clear-cut distinction in the French language between *hasard* and *fortuit*. *Hasard*, from the Arabic *az-zahr*, implies, like *alea*, dice—tractable randomness; *fortuit* is my Black Swan—the purely accidental and unforeseen. We went to the *Petit Robert* dictionary; the distinction effectively exists there. *Fortuit* seems to correspond to my epistemic opacity, *l'imprévu et non quantifiable*; *hasard* to the more ludic type of uncertainty that was proposed by the Chevalier de Méré in the early gambling literature. Remarkably, the Arabs may have introduced another word to the business of uncertainty: *rizk*, meaning property.

I repeat: Mandelbrot deals with gray swans; I deal with the Black Swan. So Mandelbrot domesticated many of my Black Swans, but not all of them, not completely. But he shows us a glimmer of hope with his method, a way to start thinking about the problems of uncertainty. You are indeed much safer if you know where the wild animals are.

* The nontechnical reader can skip from here until the end of the chapter.

* By using symmetry we could also examine the incidences below the number.

* Clearly, you do not observe 100 percent in a finite sample.

* The defense mechanism when you question their work is to say that they “do science, not philosophy” and to berate my approach of worrying about model errors. This is a common trap: people think that science is about formulating predictions (even when wrong). To me, science is about how not to be a sucker.